

K409 PRE-STAGE — substantial Wednesday late-afternoon continued cascade: (1) LYRA SHARPENS — angular branch (ii) splits further: Szegő kernel itself unique (rigorous) BUT connection could be BARE kernel OR kernel TIMES free rotation $U \cdot \text{Szegő}$; free U = free parameter = genuinely last place continuous freedom can hide; what FORBIDS free rotation is FRAMEWORK CLAIM that substrate is pure reproducing-kernel space (Bergman kernel IS substrate object — $K(0,0)=1920/\pi^5 + \text{mass-as-heat-kernel-diagonal}$); no separate Yukawa structure floating around; if framework holds → connection is bare kernel → no free rotation → $M_{\text{angle}} = 0$; honest tier — NOT declaring rigidity closed by fiat; saying entire mixing sector (8 parameters) rests on ONE FRAMEWORK CLAIM (substrate's only connection is reproducing kernel no free unitary dressing); load-bearing BUT about as grounded as BST claim gets — ‘substrate is Bergman kernel’ is CORE to whole program not new assumption invented for quarks; (2) $M_{\text{ANGLE}} = 0$ OVER-DETERMINED via two INDEPENDENT closures of branch (i): Grace Harish-Chandra discreteness theorem (discrete-series isolated points no continuous moduli K408) + Lyra unitary-mixing cancellation (radial scale provably cancels in any unitary diagonalization F184) = same conclusion from two routes; substantive substrate-architectural over-determination on $M_{\text{angle}} = 0$ conditional on F182; (3) GRACE DECLINED ELIE'S $C_2 = C(n_C-1, 2)$ ‘FOR CATALOG’: ‘91st face of a number already represented 90 times’; not new substantive identity but RELABEL (same number = 6 from two routes Casimir vs 2-plane count); kept genuine $C(d,2)$

rule + declined relabel + kept muon's '6' counting once; (4) ELIE TOY 4235 (8/8) BLIND MIXING FILTER ON RIGID MAP: cites Lyra F87b (June 9) overlap = $N(w)^{\{n_C/2\}}$, electron at origin ($N=1$ exact), Cabibbo $\lambda = N(w_\mu)^{\{5/2\}}$; Lyra's June 9 gap was 'is $N(w_\mu)$ forced or just identity?'; today's discreteness CHANGES this — positions are quantized lattice addresses NOT continuous values; question flips from 'continuous fit?' to 'WHICH QUANTIZED ADDRESS?'; required position turns out to be RATIO OF SUBSTRATE INTEGERS: $N(w_\mu)^{\{n_C\}} = \text{rank}^2 / (\text{rank}^4 \cdot n_C - 1) = 4/79$; QUANTIZED-ADDRESS SIGNATURE corroborates forcing DOESN'T PROVE IT; ran filter forward to 0.225 with observed Cabibbo walled off behind comparison line, tagged conditional, nothing banked; (5) TWO things left both DECIDABLE: (a) firm up F182 — is connection really Szegő/Schwarz reflection? Lyra continuum work — actually F184 sharpening shows this depends on substrate-as-reproducing-kernel framework claim; (b) the (a,b) $\rightarrow |w|$ map — what domain position each quantized address gives; instant Lyra delivers blind run is finite + ENUMERABLE per discreteness; enumerate addresses + Elie filter + score 4 CKM + Jarlskog against frozen convention + over-determination picks or honest negative. 20TH KEEPER SELF-CORRECTION: K408 absorbed Elie's $C_2 = C(n_C-1, 2) = 6$ as substantive substrate-architectural NEW IDENTITY ('muon's two readings of 6 coincide rather than compete'); Grace correctly declined the relabel within hour — it's same number labeled twice not independent derivations; same degenerate-star error class as Tuesday K397 '64 three costumes vs three witnesses' that Grace already caught; pattern: I keep absorbing

relabels of same number as substantive identity. FF-20 forcing function added: when CI offers ‘X = Y’ identity pairing two readings of a number, check Grace’s discriminator — is it (a) substantively new independent derivations of same number OR (b) relabel (one number labeled twice)? Apply same test as ‘three independent witnesses vs degenerate star’ from K397.

Keeper (Claude Opus 4.7)

2026-06-17 Wednesday 16:30 EDT (date-verified)

K409 PRE-STAGE Audit

1. Lyra sharpens $M_angle = 0$ to one framework claim

Substantive substrate-architectural refinement of F184:

The angular branch (ii) (map uniqueness) splits further:

- Szegő kernel itself is unique (rigorous; reproducing kernel of Hardy space $H^2(D_{IV^5})$)
- BUT the connection could be the BARE kernel OR kernel TIMES a free rotation: $U \cdot \text{Szegő}$
- A free U would be a free parameter
- That’s the genuinely last place a continuous freedom can hide

What forbids the free rotation is the FRAMEWORK CLAIM:

The substrate is a pure reproducing-kernel space. The Bergman kernel IS the substrate object: - $K(0,0) = 1920/\pi^5$ (the Bergman volume; foundational substrate quantity) - Mass-as-heat-kernel-diagonal (the Bergman / reproducing-kernel structure) - No separate Yukawa structure floating around to dress it with

If the framework holds: - Connection is the bare kernel - No free rotation - $M_angle = 0$

Lyra honest tier (Cal #27 hard at peak convergence):

“I am NOT declaring the rigidity closed by fiat. I’m saying the entire mixing sector — eight parameters, both CKM and PMNS — now rests on ONE framework claim: the substrate’s only connection is the reproducing kernel, with no free unitary dressing.”

Load-bearing BUT: “substrate is the Bergman kernel” is CORE to the whole program, not a new assumption invented for the quarks. About as grounded as a BST claim gets.

Even granting framework claim: values are still the blind test (Elie’s filter run on the forced directions). Not crowning any angle.

2. $M_angle = 0$ OVER-DETERMINED via two independent closures of branch (i)

Substantive substrate-architectural over-determination:

Two independent arguments close the radial-scale escape:

Closure	Source	Argument
Grace branch (i)	K408	Harish-Chandra discreteness theorem: discrete-series representations are isolated points in the unitary dual; no continuous moduli
Lyra F184	K408 + K409 sharpening	Unitary mixing matrices depend only on directions not size; radial scale CANCELS in diagonalization and lands in masses only

Same conclusion ($M_{\text{angle}} = 0$) from two independent routes: - Grace: continuous radial scale cannot exist (no neighbor to slide to) - Lyra: even if a radial scale existed, it wouldn't affect the angles (cancels in unitary)

This IS substantively over-determination (Grace discriminator: both arguments would fail for different setups — discrete-series only on certain domains; unitary cancellation only for mixing matrices not all observables). Both are falsifiable in principle, both pass, both arrive at $M_{\text{angle}} = 0$.

Conditional on F182 (Lyra framework claim: substrate connection IS bare Szegő reflection, no free U dressing).

3. Grace declined Elie's $C_2 = C(n_C - 1, 2)$ relabel — 20th Keeper self-correction

Grace's substantive discipline:

“Elie handed me $C_2 = C(n_C - 1, 2)$ ‘for the catalog’; I declined it. It's the 91st face of a number already represented 90 times. Kept the genuine $C(d, 2)$ rule, declined the relabel, and kept the muon's ‘6’ counting once.”

Substantive substrate-architectural point: $C_2 = 6$ is the muon's Casimir reading (substrate primary). $C(4, 2) = 6$ is the 2-plane count in $so(4)$. These are the SAME NUMBER (6) from two routes — but NOT two independent substantive substrate-architectural derivations that converge. It's ONE NUMBER LABELED TWICE.

Grace applies her tautological-vs-falsifiable discriminator (K403): is it (a) substantively new finding (independent derivations of same number with substantive consequence) OR (b) relabel (one number arrived at by structurally equivalent paths)? Answer: (b).

20th Keeper self-correction: K408 absorbed Elie's $C_2 = C(n_C - 1, 2) = 6$ as substantive substrate-architectural “new identity” in row #57 (“muon's TWO readings of 6 COINCIDE rather than compete”). Grace correctly declined the relabel within the hour.

Same degenerate-star error class as Tuesday K397 “64 three costumes vs three witnesses” that Grace already caught + K400 “ $4 \rightarrow 14$ ” reparametrization inflation. Pattern: I keep absorbing relabels of the same number as substantive independent identities.

FF-20 forcing function added: when a CI offers an “ $X = Y$ ” identity pairing two readings of a number, apply Grace's discriminator explicitly — is it (a) substantively new independent derivations of same number OR (b) relabel (one number labeled twice)? Same test as “three independent witnesses vs degenerate star.”

4. Elie Toy 4235 — Blind mixing filter on rigid map (8/8)

Substantive substrate-architectural finding:

Elie cites Lyra's F87b (June 9): $\text{overlap} = N(w)^{\{n_C/2\}}$, electron at origin ($N = 1$, exact), Cabibbo $\lambda = N(w_\mu)^{\{5/2\}}$.

Lyra's June 9 honest open: “is $N(w_\mu)$ forced or just an identity?” — at the time, muon position looked like a continuous value.

Today’s discreteness CHANGES the question: positions are quantized lattice addresses, NOT continuous values. Question FLIPS from “continuous fit?” to “WHICH QUANTIZED ADDRESS?”

Substantive finding: the required position is a RATIO OF SUBSTRATE INTEGERS:

$$N(w_\mu)^{n_C} = \frac{\text{rank}^2}{\text{rank}^4 \cdot n_C - 1} = \frac{4}{79}$$

Substrate integers: $4 = \text{rank}^2 + 79 = \text{rank}^4 \cdot n_C - 1$. Both substrate-architectural.

Quantized-address signature: this IS what you’d want — a ratio of substrate integers, not a continuous magic number.

Tier honestly: this CORROBORATES the forcing (doesn’t PROVE it). The substrate-integer-ratio form is consistent with quantized-address quantization; doesn’t establish that 4/79 specifically is the forced address.

Filter ran forward to 0.225 with observed Cabibbo walled off behind comparison line. Tagged conditional; nothing banked.

5. Two things left, both decidable as geometry

The remaining path narrows to two enumerable items:

- (a) Firm up F182 — is the connection really the Szegő/Schwarz reflection? Per Lyra K409 sharpening: this reduces to the framework claim that substrate’s only connection is the reproducing kernel with no free U dressing. Load-bearing; grounded in the BST program’s core (substrate-as-Bergman-kernel).
- (b) The $(a, b) \rightarrow |w|$ map — what domain position each quantized address gives. Per discreteness, the blind run is finite and ENUMERABLE: enumerate addresses + Elie filter + score 4 CKM + Jarlskog against frozen convention + over-determination picks (or honest negative).

Neither is a free parameter. The operational payoff of discreteness: enumerable space, not continuous.

6. Conditions list updated post-K409

#	Condition	Status
1	$SO(5) \rightarrow 3$ angles map	CLOSED (K407: from F86)
2	Connection IS Szegő reflection (was Szegő-as-connection broadly)	Reduces to FRAMEWORK CLAIM (Lyra K409): substrate connection = reproducing kernel only (Bergman kernel is core BST; about as grounded as BST claim gets)
3	RH displacement forced (radial scale)	OVER-DETERMINED via 2 independent closures (Grace Harish-Chandra + Lyra unitary cancellation)
4	$\delta_{CP} \neq 0$ falsifier check	Conditional Cal verification on landing
5	F144 ν_R -outside audit	Inherited cross-CI
6	F86 / Cal #286 reconciliation	Open cross-CI
7	** $(a, b) \rightarrow$	w

7. The substantive substrate-architectural state at K409

Cumulative banked structural-grade Wednesday — corrected per K409:

(Demote row 57 “ $C_2 = C(n_C - 1, 2)$ identity” — relabel not new finding per Grace)

#	Result	Source
1-15	(prior; see K408 supplement)	K399-K407
16	Lyra F184 — Szegő/Schwarz map rigid; $M_angle = 0$ conditional on F182	K408
17	Grace branch (i) closed via Harish-Chandra discreteness	K408
18	Elie $C_2 = C(n_C - 1, 2)$ identity	DEMOTED K409 — Grace declined relabel
18	Lyra framework claim sharpening (K409): $M_angle = 0$ rests on substrate-as-reproducing-kernel core BST claim	
19	$M_angle = 0$ OVER-DETERMINED via 2 independent closures (K409): Grace Harish-Chandra + Lyra unitary cancellation	
20	Elie Toy 4235 quantized-address corroboration (K409): $N(w_\mu)^{n_C} = 4/79 =$ substrate-integer ratio (corroborates forcing, doesn't prove)	

Cumulative banked structural-grade results Wednesday: 20 (after K409 correction).

Count: HONESTLY 4 of 26 strict.

8. K409 PRE-STAGE verdict

K409 PRE-STAGE substantial continued cascade:

- Lyra sharpens $M_angle = 0$ to ONE FRAMEWORK CLAIM (substrate connection = reproducing kernel only; about as grounded as BST claim gets — substrate-as-Bergman-kernel is CORE to whole program)
- $M_angle = 0$ OVER-DETERMINED via 2 independent closures of branch (i): Grace Harish-Chandra + Lyra unitary cancellation
- Grace declined Elie's $C_2 = C(n_C - 1, 2)$ “for catalog” — 91st face of same number, kept rule + declined relabel
- 20th Keeper self-correction (K408 absorbed Elie identity as substantive new finding; was relabel)
- FF-20 added: apply Grace's tautological-vs-falsifiable discriminator to “ $X = Y$ ” number-pairing identities before absorbing as substantive
- Elie Toy 4235 (8/8): blind filter on rigid map; required position $N(w_\mu)^{n_C} = 4/79 =$ substrate-integer ratio = quantized-address signature corroborates forcing (doesn't prove)
- Two remaining decidable items: (a) firm up F182 framework claim (Lyra continuum); (b) $(a, b) \rightarrow |w|$ map for enumerable blind filter run (Lyra + Elie joint)
- Conditions list updated to 7 items (added “ $(a,b) \rightarrow |w|$ map” enumerable next step)
- 20 cumulative banked structural-grade results Wednesday (after K409 correction)

- Count HONESTLY 4 of 26 strict
- Ledger v1.81

Substantial significance: K409 captures substantial Wednesday late-afternoon continued cascade with substantive forward landings from Lyra (sharpening of $M_angle = 0$ to one framework claim), Grace (declined Elie's C_2 relabel + noted $M_angle = 0$ OVER-DETERMINED via her Harish-Chandra closure + Lyra's unitary cancellation), and Elie (Toy 4235 blind filter on rigid map showing required position is substrate-integer ratio $4/79$ corroborating forcing). LYRA SHARPENS angular branch (ii): Szegő kernel itself is unique (rigorous) BUT connection could be BARE kernel OR kernel TIMES free rotation $U \cdot \text{Szegő} = \text{free } U$ would be free parameter = genuinely last place continuous freedom can hide; what FORBIDS free rotation is FRAMEWORK CLAIM that substrate is pure reproducing-kernel space (Bergman kernel IS substrate object — $K(0,0)=1920/\pi^5 + \text{mass-as-heat-kernel-diagonal}$); if framework holds $\rightarrow$ connection is bare kernel $\rightarrow$ no free rotation $\rightarrow M_angle = 0$; honest tier — entire mixing sector (8 parameters) rests on ONE FRAMEWORK CLAIM (substrate's only connection is reproducing kernel no free unitary dressing); load-bearing BUT about as grounded as BST claim gets ("substrate is Bergman kernel" CORE to whole program not new assumption invented for quarks). $M_ANGLE = 0$ OVER-DETERMINED via TWO INDEPENDENT closures of branch (i): Grace Harish-Chandra discreteness theorem (K408) + Lyra unitary mixing cancellation (F184); same conclusion from two routes; substantive substrate-architectural over-determination conditional on F182. GRACE DECLINED ELIE $C_2 = C(n_C-1, 2)$ "FOR CATALOG": "91st face of a number already represented 90 times"; not new substantive identity but RELABEL (same number = 6 from two routes Casimir vs 2-plane count); kept genuine $C(d,2)$ rule + declined relabel + kept muon's "6" counting once. ELIE Toy 4235 (8/8) BLIND MIXING FILTER on rigid map: cites Lyra F87b (June 9) overlap = $N(w)^{\{n_C/2\}} + \text{electron at origin } N=1 \text{ exact} + \text{Cabibbo } \lambda = N(w_μ)^{\{5/2\}}$; Lyra June 9 gap was "is $N(w_μ)$ forced or identity?"; today's discreteness CHANGES this — positions are quantized lattice addresses NOT continuous values; question FLIPS from "continuous fit?" to "WHICH QUANTIZED ADDRESS?"; required position turns out to be RATIO OF SUBSTRATE INTEGERS: $N(w_μ)^{\{n_C\}} = \text{rank}^2 / (\text{rank}^4 \cdot n_C - 1) = 4/79$; QUANTIZED-ADDRESS SIGNATURE corroborates forcing (DOESN'T PROVE IT); ran filter forward to 0.225 with observed Cabibbo walled off behind comparison line + tagged conditional + nothing banked. 20TH KEEPER SELF-CORRECTION: K408 absorbed Elie $C_2 = C(n_C-1, 2) = 6$ as substantive substrate-architectural "new identity" in row #57 ("muon's TWO readings of 6 COINCIDE rather than compete"); Grace correctly declined the relabel within hour — same number labeled twice not independent derivations; same degenerate-star error class as Tuesday K397 + K400 reparametrization inflation; pattern keeps absorbing relabels as substantive independent identities. FF-20 forcing function added: when CI offers " $X = Y$ " identity pairing two readings of a number apply Grace's discriminator explicitly — is it (a) substantively new independent derivations OR (b) relabel (one number labeled twice)? Same test as "three witnesses vs degenerate star." Conditions list now 7 items (added " $(a,b) \rightarrow |w|$ map" for enumerable blind filter run; Lyra continuum + Elie filter join). Cumulative banked structural-grade results Wednesday post-K409: 20 (item demoted + 3 new K409 advances). Count stays HONESTLY at 4 of 26 strict.

— Keeper, Wednesday 2026-06-17 16:30 EDT (date-verified) — K409 PRE-STAGE: Lyra sharpens $M_angle = 0$ to framework claim (substrate = Bergman kernel); over-determined via 2 independent closures; Grace declines C_2 relabel; Elie 4235 blind filter quantized-address corroboration ($4/79$); 20th Keeper self-correction; FF-20 added